

Def. Let $f: X \rightarrow Y$ be a map. Define a graph of f as:

$$G_f \stackrel{\text{def}}{=} \{(x, f(x)) \mid x \in X\} \subseteq X \times Y.$$

Thm. Let $I: X \rightarrow X: x \mapsto x$ be an Identity map.

X is Hausdorff $\Leftrightarrow G_I$ is closed.